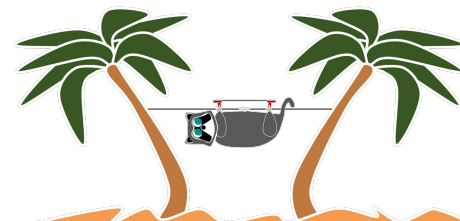


Collaboration in Multi-Agent Systems

Riwa Karam

ARM External Talk

Tuesday, September 15th, 2026

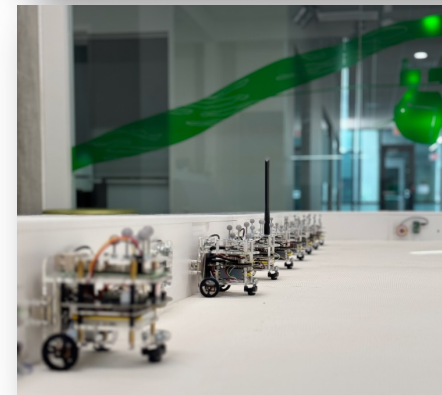
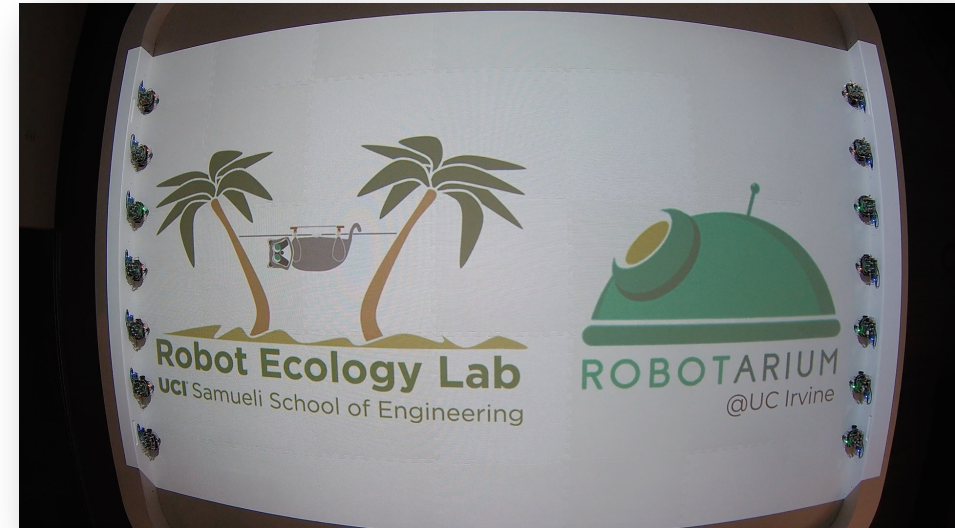


Robot Ecology Lab
UCI Samueli School of Engineering

A bit about me

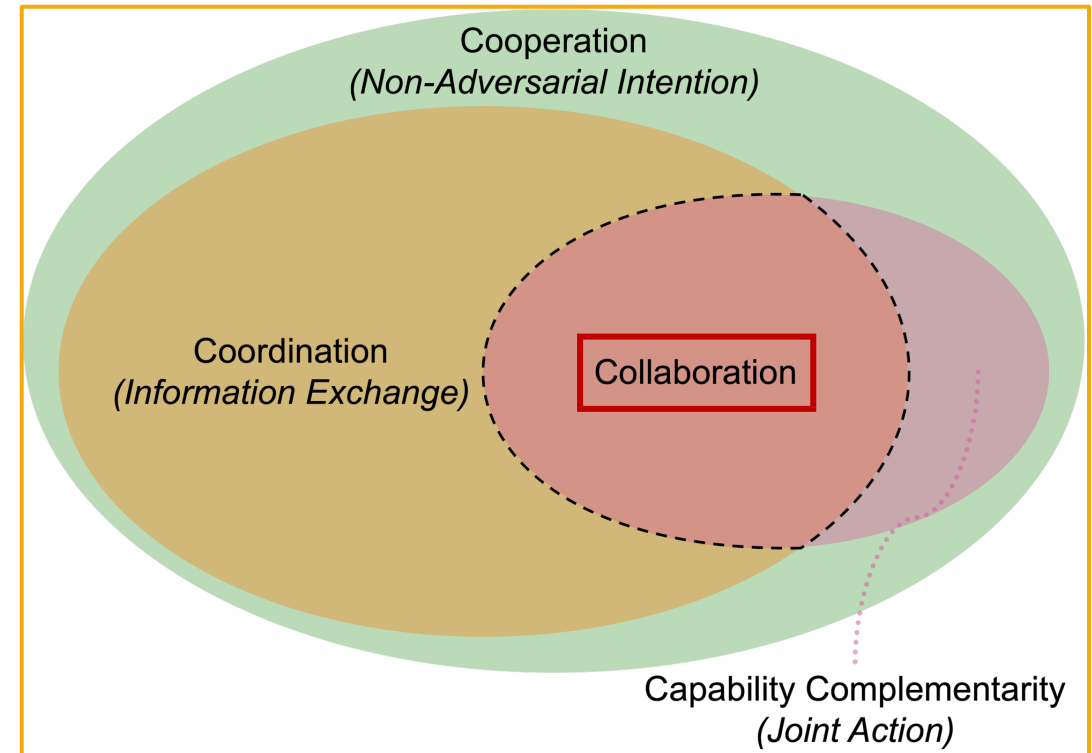
- PhD in Computer & Electrical Engineering, University of California, Irvine (Sep 2023 – Expected Jun 2027)
 - Advisors: Professor Magnus Egerstedt and Professor Yanning Shen
 - Research Lab: UCI Robot Ecology Lab
- M.S. in Computer & Electrical Engineering, UCI (Sep 2023 – Jun 2025)
- B.S. in Computer Engineering, University of Balamand, Al-Kurah, Lebanon (Aug 2020 – Jun 2023)
- U.S. industry experience:
 - Summer 2024: Mechatronics and Control Systems Engineer Intern @ ASML
 - Summer 2025: Mechatronics Research Engineer Intern @ ASML

A bit about the UCI Robot Ecology Lab



Collaboration in Multi-Agent Systems

- What is *collaboration* in multi-agent systems?
- How is it different than *cooperation* and *coordination*?



Collaboration in Multi-Agent Systems

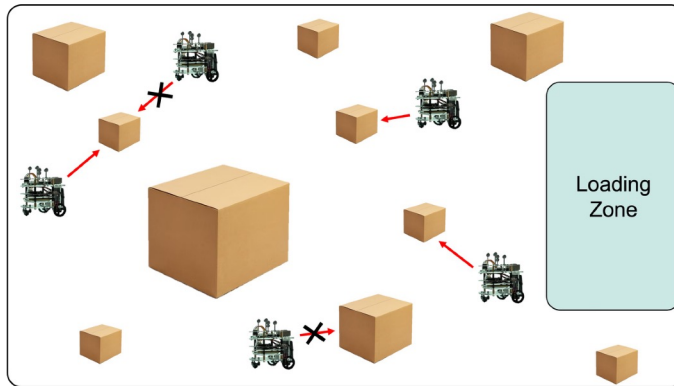


The Avengers fight Loki's army in NYC [1]

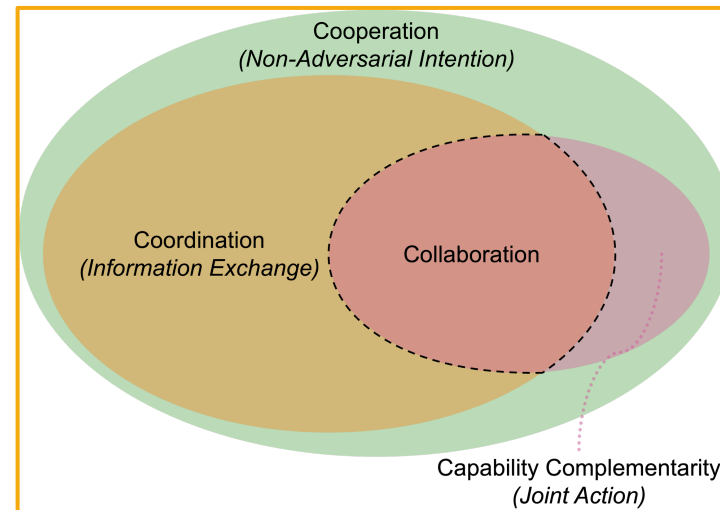
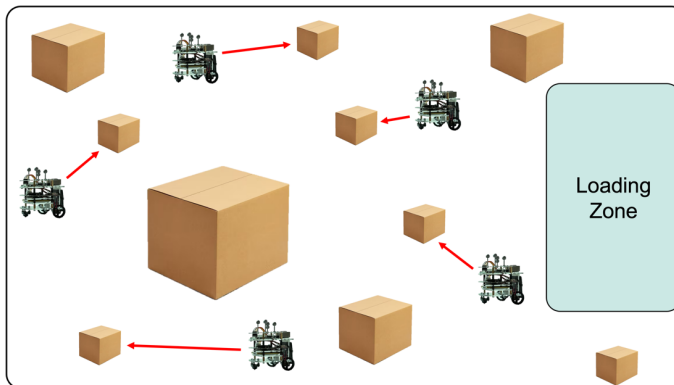
- Multi-hero system
- Each hero brings something to the table (power or ability)
- They work together and combine their abilities to complete their mission

Collaboration in Multi-Agent Systems

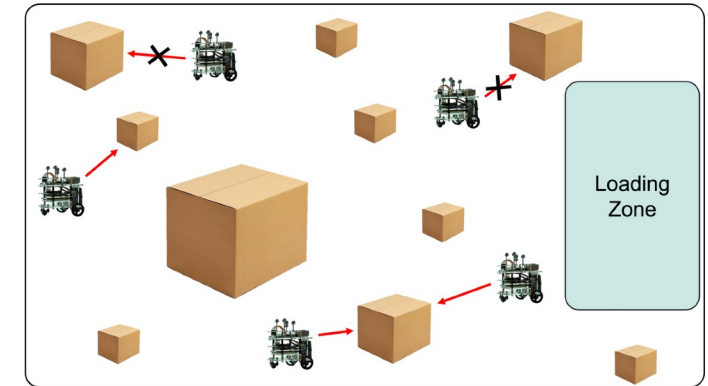
Cooperation



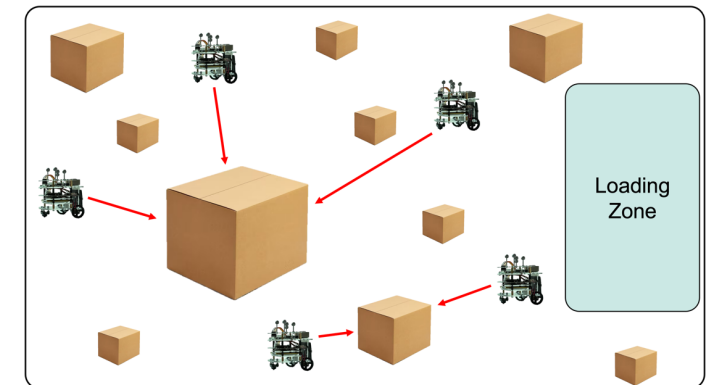
Coordination



Capability Complementarity



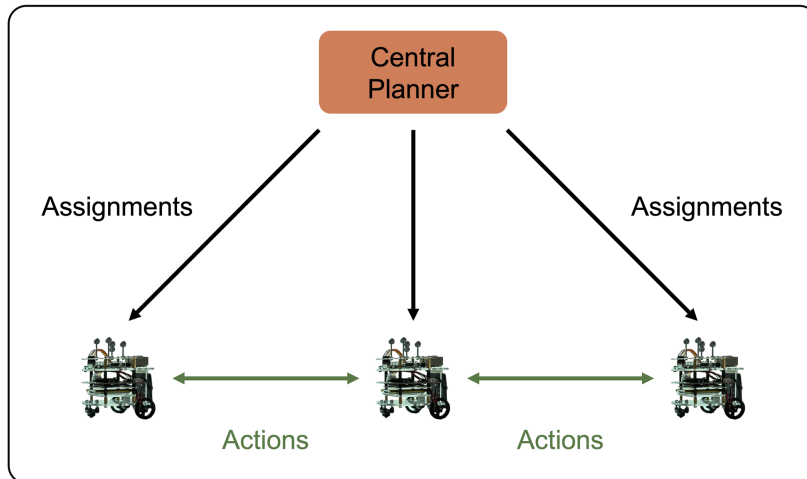
Collaboration



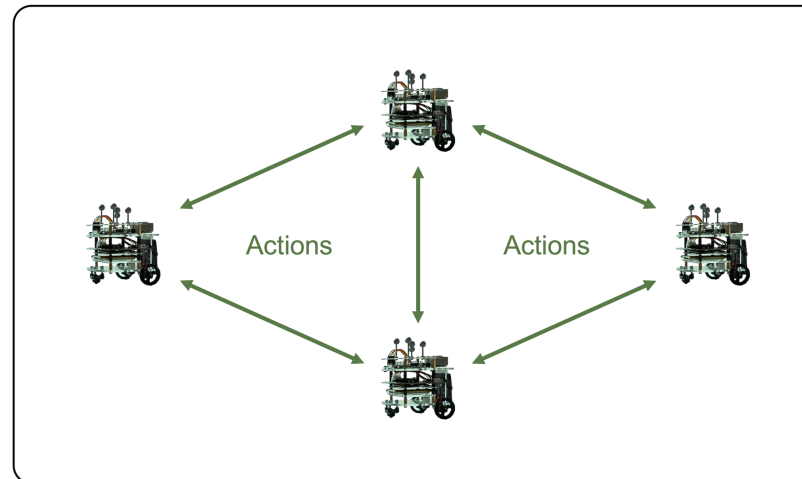
Collaboration in Multi-Agent Systems

Organizational Structures in Multi-Agent Systems

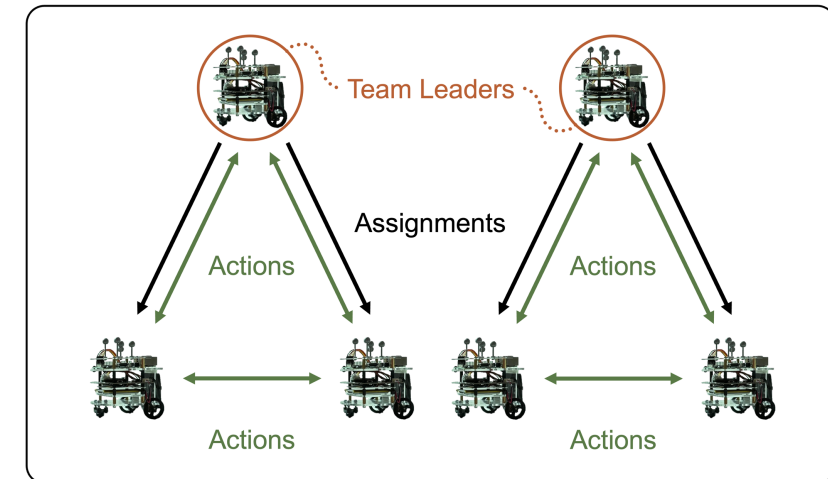
Centralized



Decentralized



Hierarchical



Collaboration in Multi-Agent Systems

Papers	Centralized	Decentralized	Hierarchical	Ecological	HSI	Game Theoretic	Learning-Based
[7, 9, 104, 105]	×	✓	×	×	×	×	✓
[8, 57, 77]	✓	×	×	✓	×	×	×
[30]	×	✓	×	×	×	×	✓
[58, 59]	✓	×	×	×	×	×	×
[60]	✓	×	×	×	×	×	✓
[61, 76]	×	✓	×	✓	×	×	×
[62, 63, 94, 95, 99, 100]	×	✓	×	×	×	✓	×
[64, 65]	×	×	✓	×	×	×	✓
[66]	×	×	✓	×	×	×	×
[67, 68, 89, 90]	×	✓	×	×	✓	×	×
[78]	×	✓	×	✓	×	×	✓
[87, 91]	×	×	✓	×	✓	×	×
[88]	×	×	✓	×	✓	×	✓

Why decentralized?

- Scalability
- Robustness

Research Questions

- Why should we care about collaboration in multi-agent systems? How can it be beneficial in real-world scenarios? Is it beneficial at all?
- Can we look at nature and get inspired by how different species co-exist and *collaborate* to survive?

Real-World Scenario



Observations:

- 4 Distinct Regions of Interest
- Each has a certain fire intensity
- Each has a certain population density

Multi-Robotic Solution?

Los Angeles County fires: Maps of where the wildfires are [1]

Ecological Inspiration: Decision Making

Lioness adopts deceased sister's cubs [1]



Hamilton's Rule of Altruism: [2]

$$rB \geq C$$

- r : genetic relatedness
- B : benefit to the recipient
- C : cost to the donor

- Genetic relatedness: sibling relationship, lioness and her deceased sister
- Benefit to the lion cubs' deceased mother: her genes will survive through her cubs' survival
- Cost to the lioness: sharing her lion cubs' resources and splitting her care.

How to translate Hamilton's rule to robotic systems?

Lioness

Team of Robots

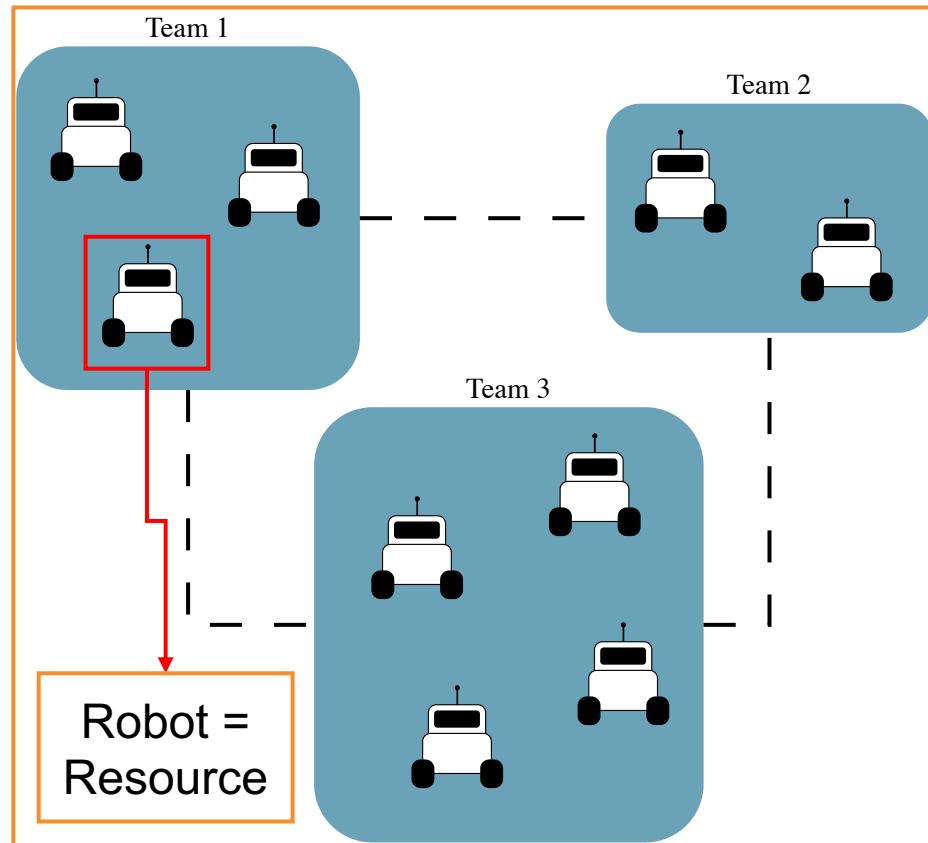
Genetic Relatedness

Mission Weights

[1]: <https://www.dailymail.co.uk/news/article-4418844/Lioness-suckles-orphaned-cubs-adopted-sister.html>

[2]: W. D. Hamilton, "The genetical evolution of social behaviour. ii," Journal of Theoretical Biology, vol. 7, no. 1, pp. 17–52, 1964.

Resource Allocation in Multi-Team Systems



Graph Modeling: [1]

- Let there be m teams of robots
- $G = (V, E)$: undirected graph
- $V = \{v_1, \dots, v_m\}$: set of **teams**
- $E \subseteq V \times V$: set of **potential interactions**

Key Idea:

Compute potential interactions inspired by Hamilton's Rule [2]

Assumption 1:

All agents in all m teams are homogenous, i.e., identical

Resource Allocation in Multi-Team Systems

Mission Evaluation Function for team k : $F_k(n_k)$, with n_k : number of robots in team k

Assumption 2:

For each team k , the mission evaluation function $F_k(n_k) : \mathbb{N} \rightarrow \mathbb{R}$ satisfies the following properties:

1. **Strictly Increasing:** For all $n \geq 1$,

$$F_k(n+1) > F_k(n)$$

2. **Discrete Concavity:** For all $n \geq 2$,

$$F_k(n+1) - F_k(n) \leq F_k(n) - F_k(n-1)$$

Higher is better

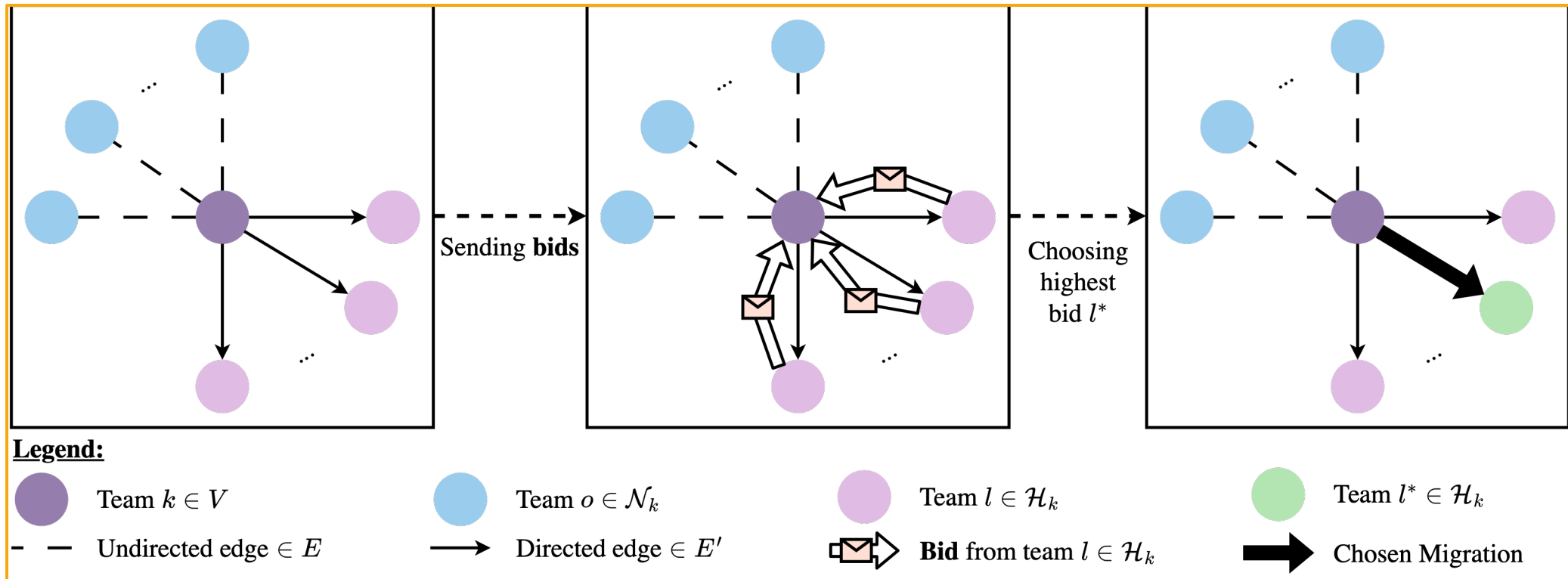
Diminishing Returns

Resource Allocation in Multi-Team Systems

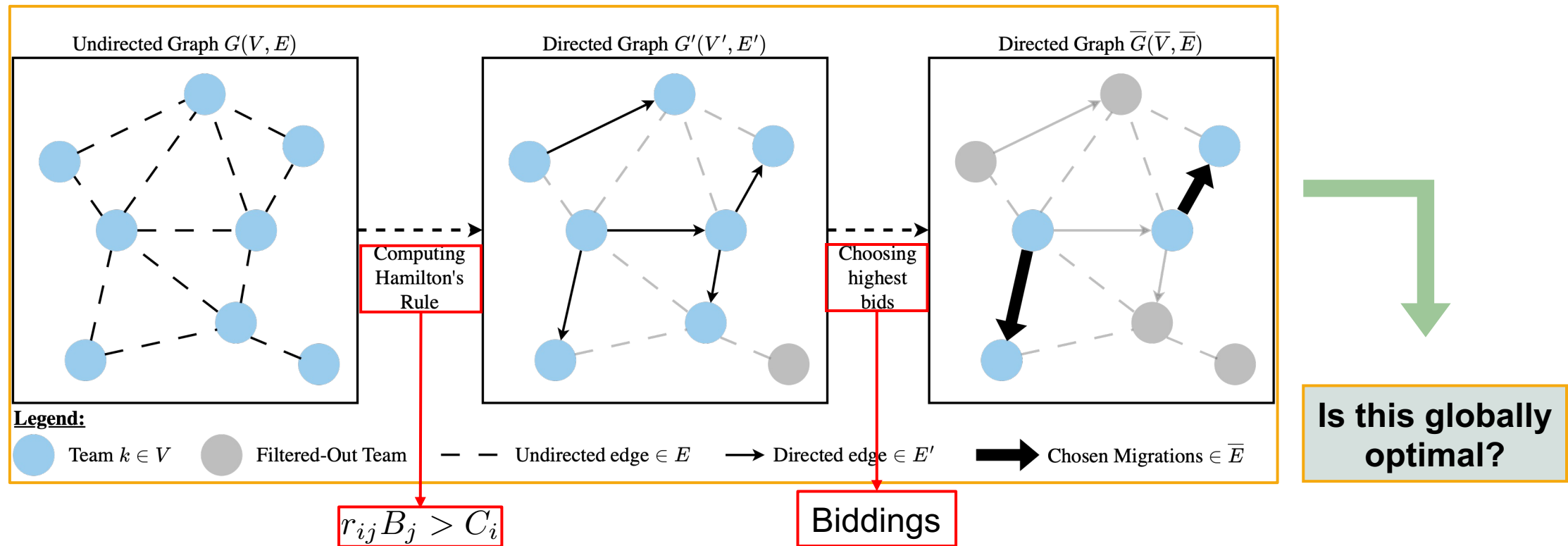
Hamilton's Rule [1]	Hamilton's Rule for Team-Level Resource Allocation	
$rB \geq C$	$r_{ij}B_j > C_i, \quad (i, j) \in E$	
r : genetic relatedness	r_{ij} : relative mission importance	$r_{ij} = \frac{w_j}{w_i} \quad \text{and} \quad r_{ji} = \frac{w_i}{w_j} = \frac{1}{r_{ij}}$
B : benefit to the recipient	B_j : benefit to team j	$B_k = F_k(n_k + 1) - F_k(n_k)$
C : cost to the donor	C_i : cost to team i	$C_k = F_k(n_k) - F_k(n_k - 1)$

Theorem 1. Consider two teams i and j , with $i, j \in V$, each with at least one agent. Under Assumptions 1 and 2, if $r_{ij}B_j > C_i$, it follows that $r_{ji}B_i \leq C_j$.

Resource Allocation in Multi-Team Systems



Resource Allocation in Multi-Team Systems



Resource Allocation in Multi-Team Systems

Global Mission Evaluation Function:

$$\mathcal{G}(n_1, \dots, n_m) = \sum_{k=1}^m w_k F_k(n_k)$$

N total number of robots across the m teams

Global Inequality Rule:

$$\mathcal{G}(n'_1, \dots, n'_m) > \mathcal{G}(n_1, \dots, n_m)$$



Global benefit instead of local (or greedy)

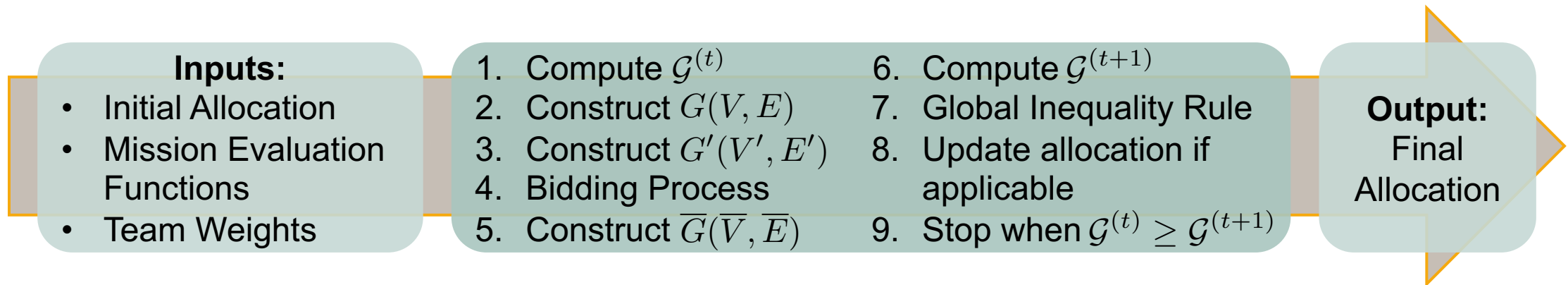
$$\Omega = \left\{ (n_1, \dots, n_m) \in \mathbb{Z}_{\geq 0}^m \mid \sum_{k=1}^m n_k = N \right\} : \text{set of all allocations of the } N \text{ agents between the } m \text{ teams}$$

Resource Allocation in Multi-Team Systems

Lemma 1. There exists an allocation $(n_1^*, \dots, n_m^*) \in \Omega$, such that

$$\mathcal{G}(n_1^*, \dots, n_m^*) \geq \mathcal{G}(n_1, \dots, n_m) \quad \forall (n_1, \dots, n_m) \in \Omega$$

In other words, $\mathcal{G}$ attains a maximum value on Ω , and $(n_1^*, \dots, n_m^*)$ is an optimal allocation.



Theorem 2. This iterative process terminates in a finite number of steps and converges to an allocation $(n_1^*, \dots, n_m^*) \in \Omega$ that maximizes $\mathcal{G}$.

Resource Allocation in Multi-Team Systems

Team Mission



Optimal Area Coverage



Voronoi-Based Coverage Control [1]

$N \in \mathbb{Z}^+$ Robots



Convex and Compact $\mathcal{D} \subset \mathbb{R}^d$



$$L(p) = \sum_{i \in \mathcal{N}} \int_{\mathcal{V}_i} \|p_i - q\|^2 \phi(q) dq$$



Dynamic Area
Coverage with Fire-
Fighting Robots [2]

Centroidal Voronoi Tessellation (CVT)

$$p_i = c_i = \frac{\int_{\mathcal{V}_i} q \phi(q) dq}{\int_{\mathcal{V}_i} \phi(q) dq}, \quad \forall i \in \mathcal{N}$$

- Centroid: $c_i \in \mathbb{R}^d$
- CVT: $c = [c_1^\top, c_2^\top, \dots, c_N^\top]^\top$

Resource Allocation in Multi-Team Systems

Mission Evaluation Function: $F(N) = -L(N, c)$, with: $L(N, p) = \sum_{i \in \mathcal{N}} \int_{\mathcal{V}_i} \|p_i - q\|^2 \phi(q) dq$

Assumption 2:
Mission Evaluation Function Properties

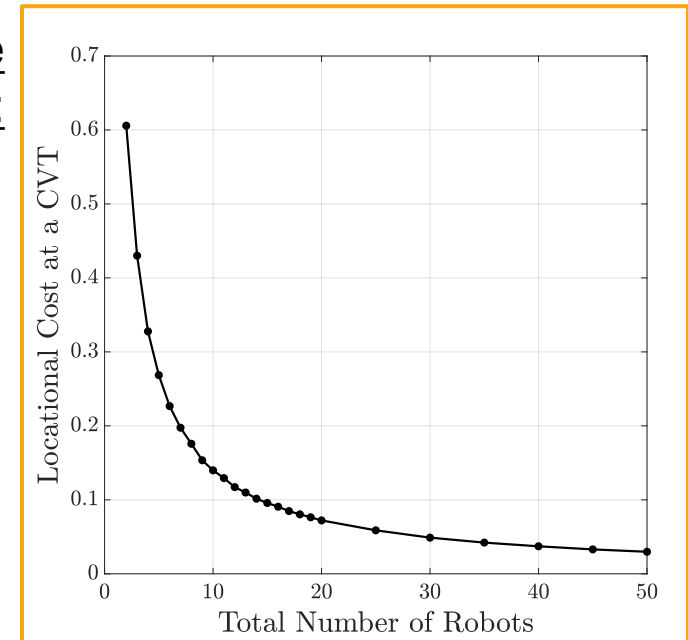
Discrete
Convexity:

Strictly Decreasing:

Theorem 3. Suppose $p_i \in \mathcal{D}$, $\forall i \in \mathcal{N}$. By adding one more robot $p_{N+1} \in \mathcal{D} \setminus \{p_i\}$, $\forall i \in \mathcal{N}$, the locational cost decreases, i.e.,

$$L(N + 1, p') < L(N, p), \quad \forall N \in \mathbb{Z}^+$$

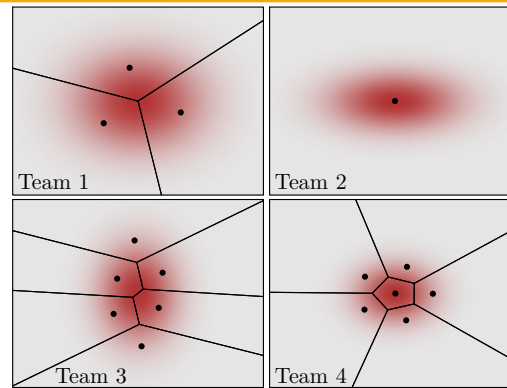
where $p' = [p^\top, p_{N+1}^\top]^\top$.



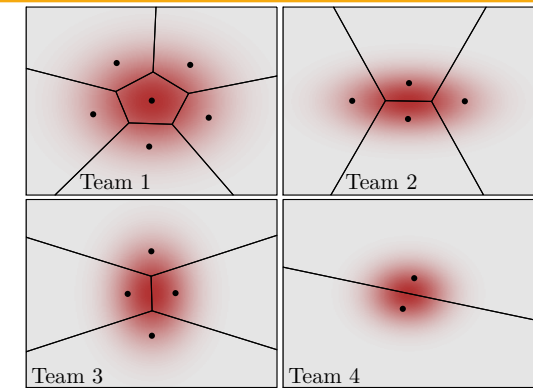
Resource Allocation in Multi-Team Systems

Case 1:

- **Same** mission importances
- **Different** density functions



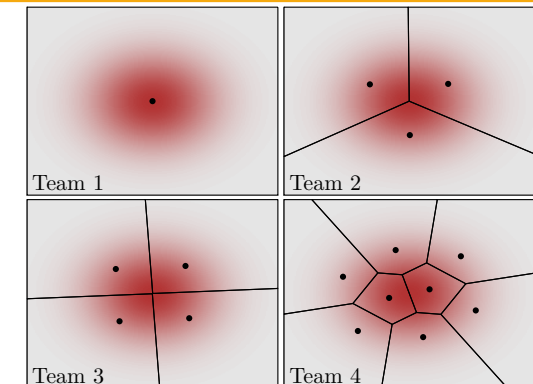
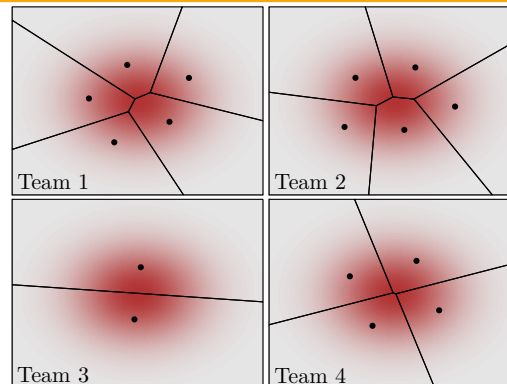
Initial Allocation



Final Allocation

Case 2:

- **Different** mission importances
- **Same** density functions



Resource Allocation in Multi-Team Systems

- Ecologically-Inspired Framework: Hamilton's Rule
- Multi-Team Environments
- Optimal Resource Allocation
- Robots Considered as Resources



Framework Achieves Optimal Performance for the Entire System through Altruistic Behavior

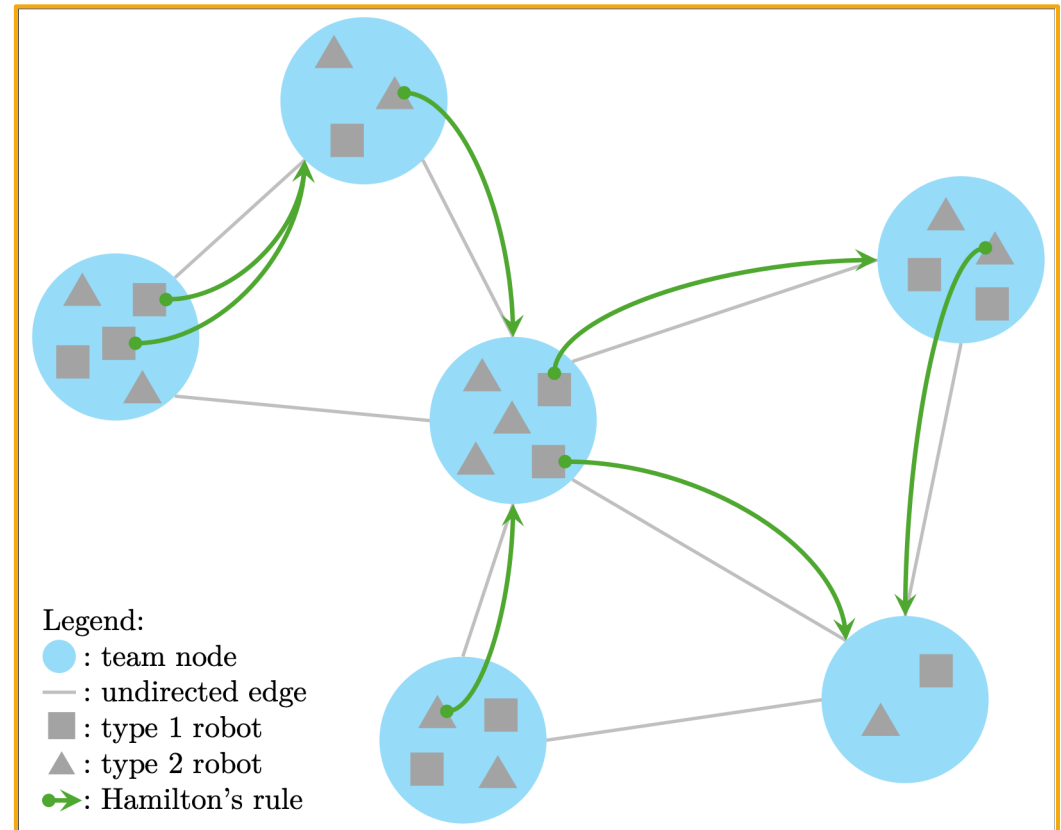
BUT!

- Assumptions on $F_k(n_k)$
- Homogeneous robots
- No migration cost
- Centralized
- Average case complexity: $O(|V||E|)$
- Worst case complexity: $O\left(|E|\binom{N-1}{M-1}\right)$
- And what about *collaboration*?

Learning Heterogeneous Altruistic Collaboration



Kenyan lioness adopts 6 Beisa Oryxes:
A heterogeneous altruism [1]



Learning Heterogeneous Altruistic Collaboration

Heterogeneous Set



$$\mathcal{S}_v = \{r \in \mathcal{R} \mid x_{r,v} = 1\}, \mathcal{S}_v \subset \mathcal{R}.$$

Heterogeneous Capabilities



$$\kappa_r \in \{0, 1\}^Q$$

Mission Evaluation Function



$$\mathcal{F}_v : \{\mathcal{S} \mid \mathcal{S} \subseteq \mathcal{R}\} \rightarrow \mathbb{R}$$

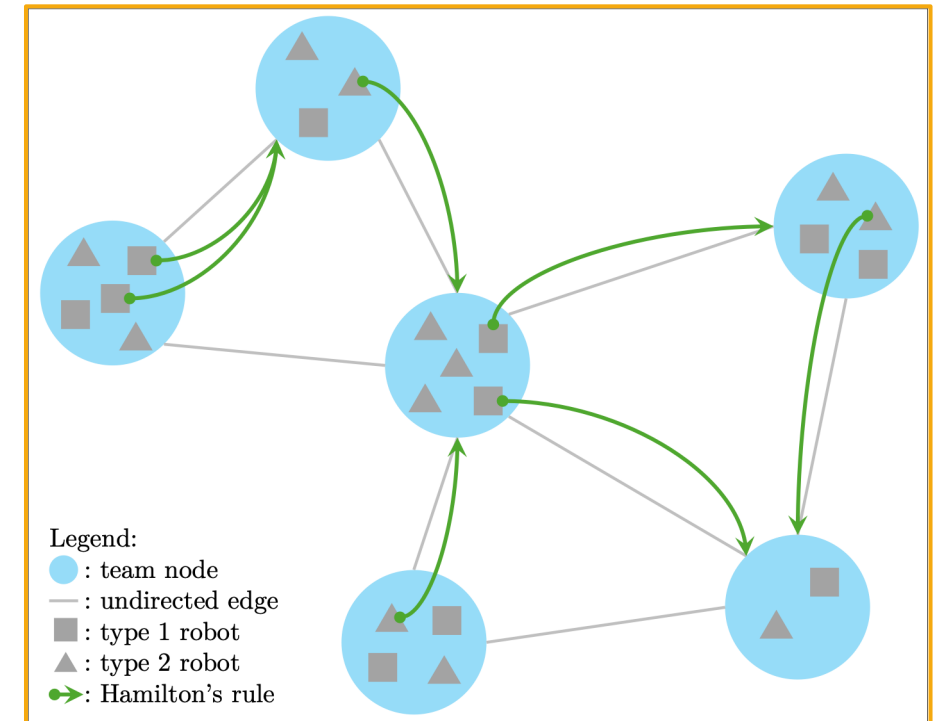
max global performance
s.t. assignment constraints
mission constraints
...



max global performance - global cost
s.t. assignment constraints
mission constraints
...

Learning Heterogeneous Altruistic Collaboration

Hamilton's Rule [1]	Hamilton's Rule for Team-Level Heterogeneous Resource Allocation
$rB \geq C$	$r_{ij} B_{r,j}(\mathcal{S}_j) > C_{r,i}(\mathcal{S}_i)$
r	$r_{ij} = \frac{w_j}{w_i} \text{ and } r_{ji} = \frac{w_i}{w_j} = \frac{1}{r_{ji}}$
B	$B_{r,j}(\mathcal{S}_j) = \mathcal{F}_j(\mathcal{S}_j \cup \{r\}) - \mathcal{F}_j(\mathcal{S}_j)$
C	$C_{r,i}(\mathcal{S}_i) = \mathcal{F}_i(\mathcal{S}_i) - \mathcal{F}_i(\mathcal{S}_i \setminus \{r\})$



The heterogeneous altruistic resource allocation problem is shown to be an NP-Hard problem

Learning Heterogeneous Altruistic Collaboration

$\max$ global performance - global cost
 s.t. assignment constraints
 mission constraints
 ...



$\max_{X^{k+1}} \mathcal{G}(X^{k+1}) - \lambda \mathcal{C}(X^k, X^{k+1})$
 s.t. assignment constraints,
 Hamilton's rule constraints
 mission-specific constraints,
 ...

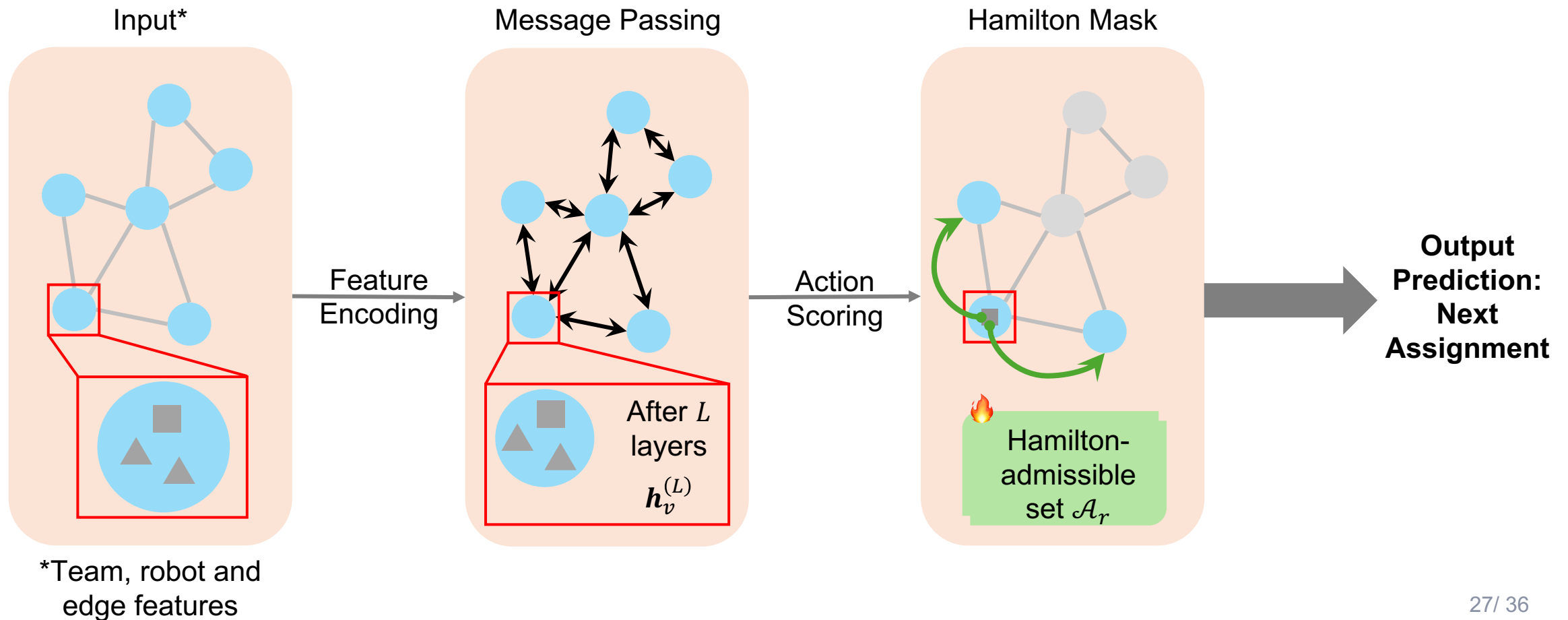
Tractability and scalability problem: Solving an NP-hard problem at each iteration of the algorithm!

Learning-Based Solution?

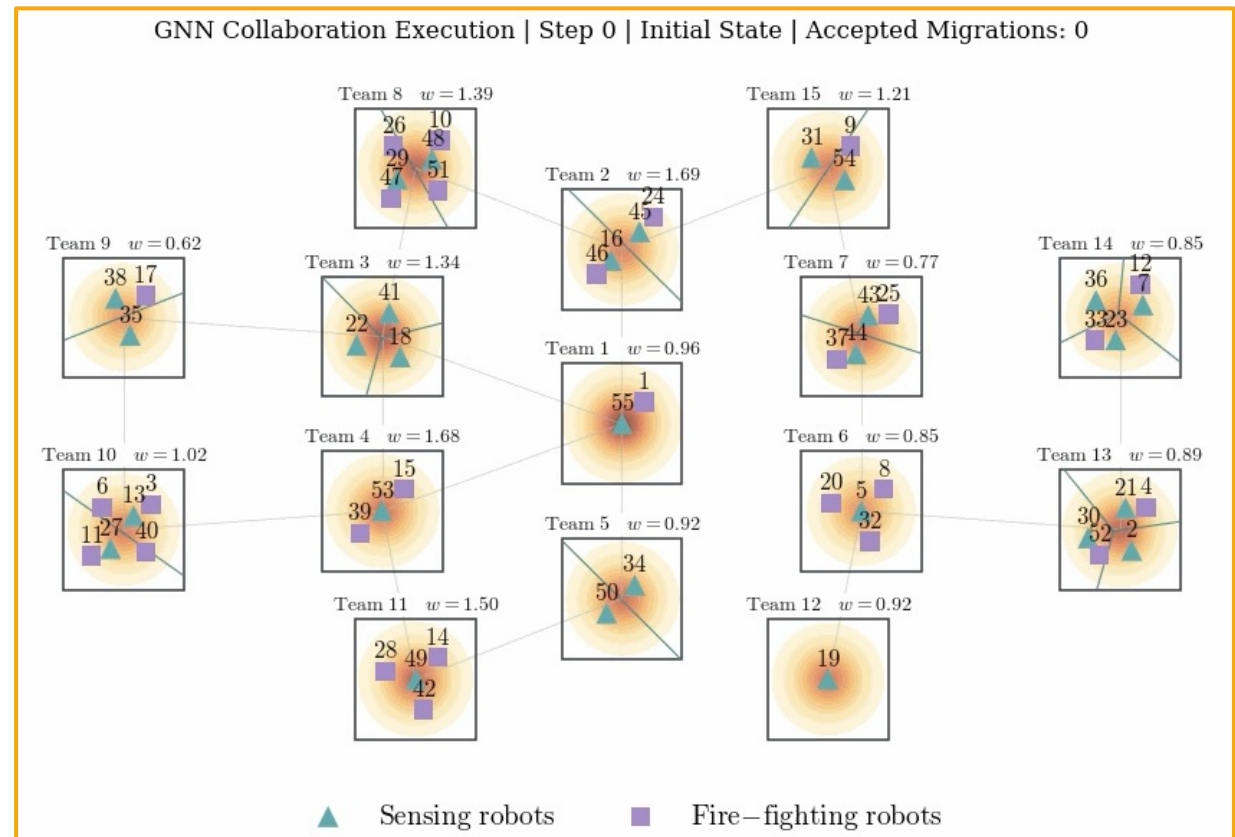
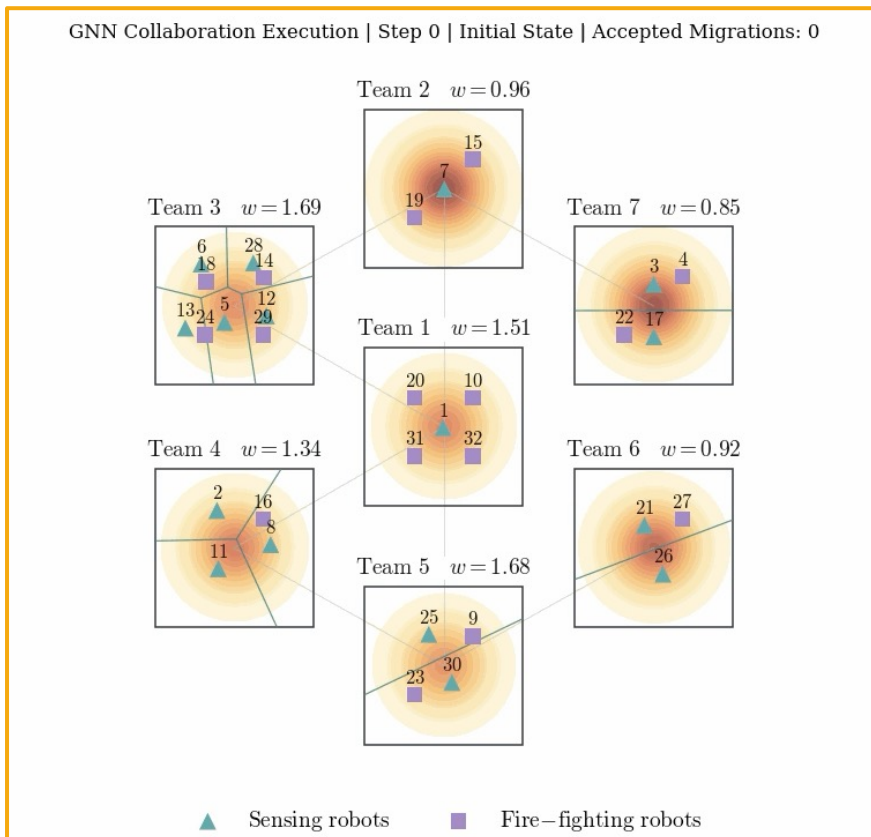


Graph Neural Network policy to approximate the solution of the problem at each iteration

Learning Heterogeneous Altruistic Collaboration



Learning Heterogeneous Altruistic Collaboration



Learning Heterogeneous Altruistic Collaboration



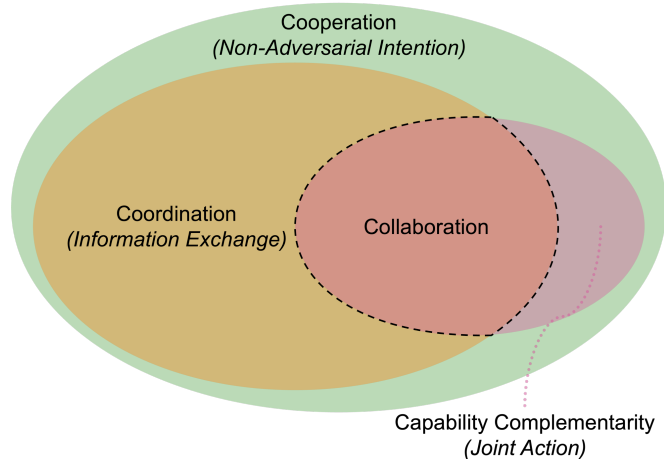
Learning Heterogeneous Altruistic Collaboration

RUNTIME SCALING OF THE EXACT OPTIMIZATION AND THE GNN POLICY. EXACT AND GNN TOTAL RUNTIMES ARE FULL ITERATIVE EXECUTION RUNTIMES. MEAN STEP RUNTIMES ARE COMPUTED FROM THE PER-STEP RUNTIME EXECUTION.

Teams	Robots	Opt. Total (s)	Opt. Mean Step (s)	Opt. Steps	GNN Total (s)	GNN Mean Step (s)	GNN Steps
3	9	0.051	0.025	2	0.048	0.012	4
4	12	0.090	0.030	3	0.012	0.006	2
5	15	0.781	0.390	2	0.009	0.005	2
6	18	35.4	11.8	3	0.170	0.034	5
7	21	∞	—	1	0.201	0.050	4
8	24	∞	—	—	0.298	0.075	4
9	27	∞	—	—	0.816	0.204	4
10	30	∞	—	—	0.658	0.219	3
15	45	∞	—	—	1.957	0.326	6
25	75	∞	—	—	2.787	0.697	4
35	105	∞	—	—	16.3	2.335	7
40	120	∞	—	—	36.0	2.001	18
45	135	∞	—	—	48.5	2.425	20
50	150	∞	—	—	55.3	2.767	20

Some Concluding Remarks

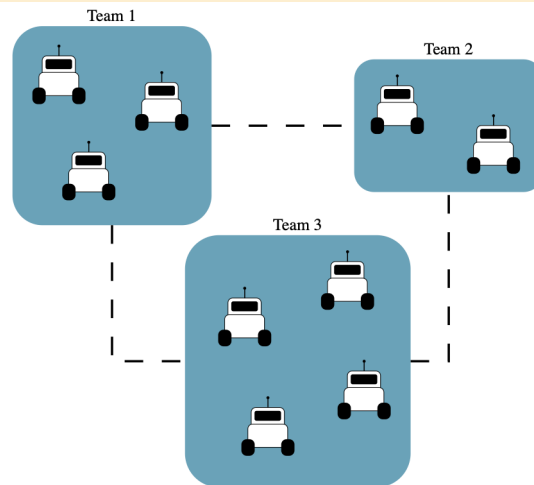
Collaboration in Multi-Agent Systems



Collaboration?

- Allows for joint action
- Expands the collective action and feasible sets

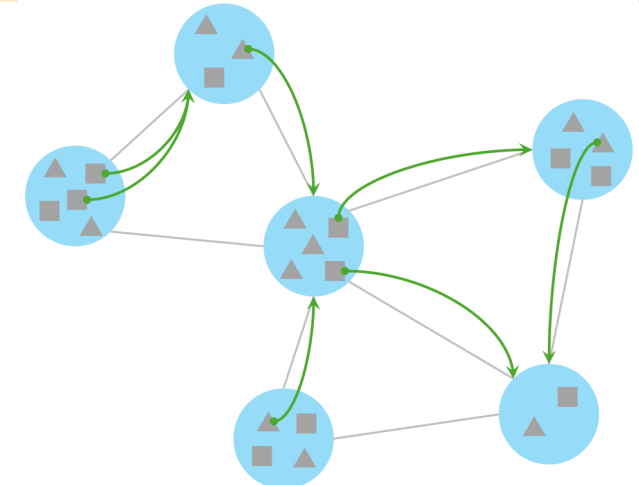
Resource Allocation in Multi-Team Systems



Resource assignment?

- Multi-team setting with robots as resources
- Altruistic decision-making

Learning Altruistic Collaboration



Learning-based methods?

- Real-time approximation of the intractable NP-hard problem

Ongoing Research Interests and Questions

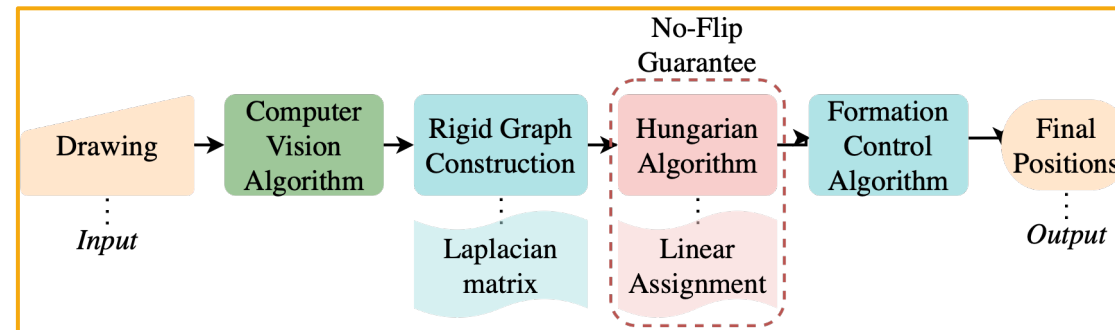
- How can we leverage machine learning to approximate long-horizon collaborative task scheduling?
- Can robots collaborate to optimally and safely complete tasks with energy-efficient mobility?
- Can robots learn to safely collaborate online in new and unexplored environments?
- How can human-swarm collaboration become more accessible to non-expert end-users?

MPC*-Inspired Collaborative Robot Mobility



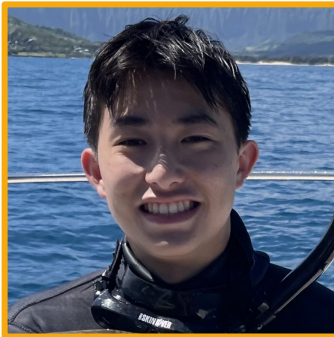
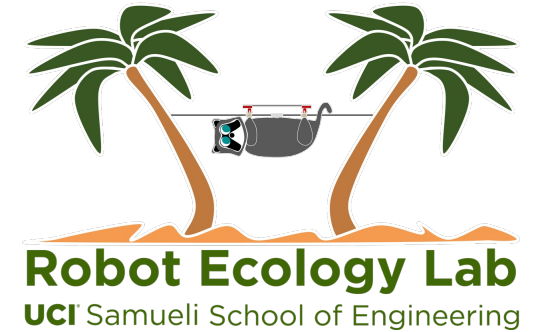
- Research Idea: Energy-based collaborative scheduling using machine learning
- Motivation: Formulated mixed-integer linear program becomes intractable at scale
- Expected Contribution: Online approximation of the long-horizon collaborative scheduling for energy-efficient mobility

Human-Swarm Collaboration

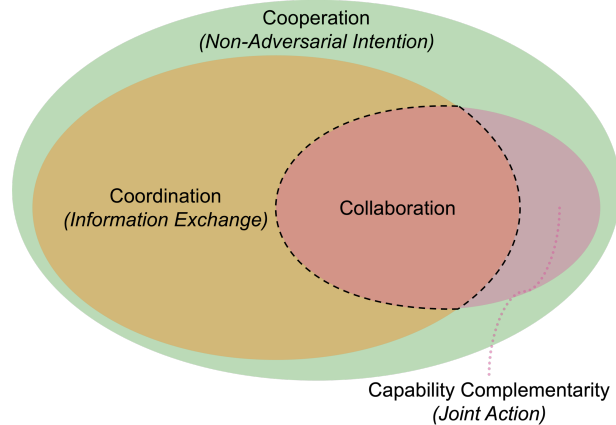


- Research Idea: User-friendly graphical user interface that allows non-expert users to specify and adapt swarm formations through freehand sketches
- Motivation: Accessibility and usability of human-swarm collaboration
- Expected Contribution: User study quantifying usability of the interface

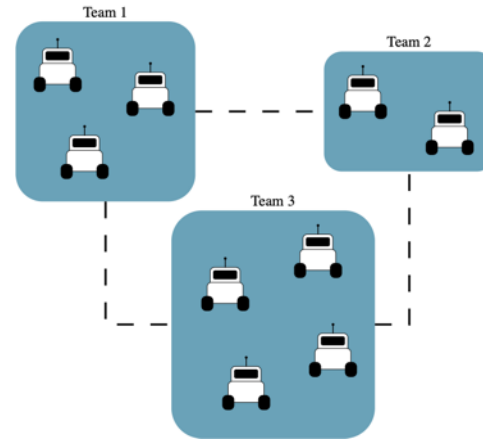
Acknowledgments



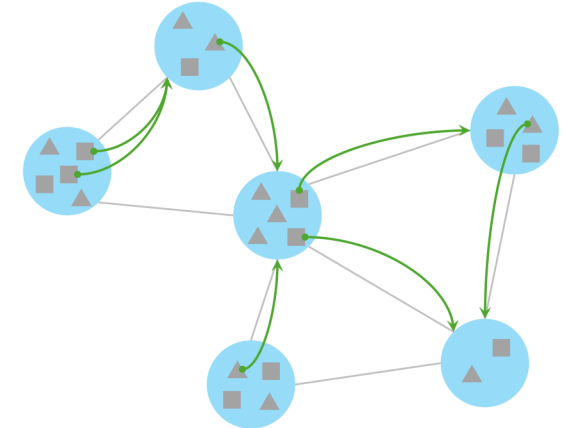
Collaboration in Multi-Agent Systems [1]



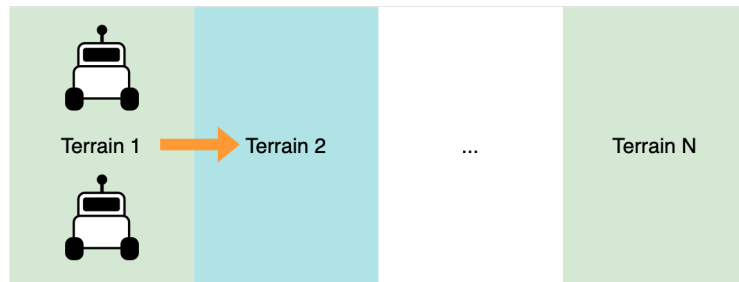
Resource Allocation in Multi-Team Systems [2]



Learning Altruistic Collaboration [3]

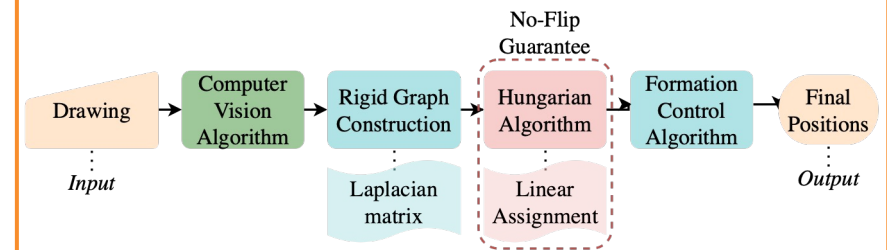


MPC-Inspired Collaborative Robot Mobility



Thank you!

Human-Swarm Collaboration [4,5]



- [1]: **Karam, R.**, Nguyen, A. A., Lin, R., Martin, D. R., Morales, D., Butler, B. A., & Egerstedt, M., "Collaboration in Multi-Robot Systems: Taxonomy and Survey over Frameworks for Collaboration." (under review in IEEE CSM, 2026)
- [2]: **Karam, R.**, Lin, R., Butler, B. A., & Egerstedt, M., "Resource Allocation with Multi-Team Collaboration based on Hamilton's Rule," in Proceedings of the IEEE Conference on Decision and Control (CDC) (pp. 6891-6898), 2025, IEEE.
- [3]: **Karam, R.**, Lin, R., Butler, B. A., & Egerstedt, M., "Learning Altruistic Collaboration in Heterogeneous Multi-Team Systems." (under review in IEEE T-RO, 2026)
- [4]: **Karam, R.**, "A Graphical Interface for Specifying and Establishing Multi-Robot Formations." Master's thesis, University of California, Irvine, 2025.
- [5]: **Karam, R.**, Kuo, I., & Egerstedt, M., "Freehand Sketching for End-User Programming of Robot Swarms." (submitted, 2026)